Q) Coal Assignment #1:-
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Roll-no:- 24K-0737.

Q#1,-

(a) memory-map:-

Extra segment
Code segment
Data segment
Stack-segment
Other unused/reserved

physical address = Data-segment x 10h + offset.

(b)(i) 1234:5678.

Sol

1234 x 10h + 5678.

12340 + 5678.

= 179B8h. (No wraps around)

(ii) FFFF:000F

Sol:-

FFFF x 10h + 000F.

$$\begin{array}{r} FFFF0 \\ +\ 000F \\ \hline FFFFF \end{array}$$

= FFFFFh (No wrap around)

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i) if the bit exceeds 20 bits then its a wrap around.

(iii) F000 : FFF0

Sol.

F000 x 10h + FFF0

$$
\begin{array}{r}
F0000 \\
+\ FFF0 \\
\hline
FFFF0
\end{array}
$$

∴ FFFF0h (No wrap around)

(iv) F800 : 9000

Sol.

F800 x 10h + Offset.

F800 x 10h + 9000.

$$
\begin{array}{r}
F8000 \\
+\ 9000 \\
\hline
101000
\end{array}
$$

101000h

25 bits

(wraps around)

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Q#2:

· General purpose Registers:

EAx : arthimatic accumulator Ex: ADD EAx, EBx
EBx : Base register for memory access.
ECx : loop counter.
EDx : used with eax in division

· Index / base / stack / instruction:-

ESI : Source for string copy.
EDI : Destination for string copy.
ESP : Point to stack pointer.
EBP : Access function Arguments.
EIP : point to read instruction.

Q#3:

0x1234ABCD into memory starting at address 4000h.

Little-endian		Big-endian	
4000h	CD	4000h	12
4001h	AB	4001h	34
4002h	34	4002h	AB
4003h	12	4003h	CD

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Q#4:

mov Ax, 8F7Ah   Ax=? CF=? SF=?
ADD Ax 7AF8h   Ax=? CF=?, SF=?
mov Bx, 0FA77h   Bx=?, CF=? SF=?
INC Bx   Bx=? CF=? SF=?

Sol:

mov Ax, 8F7Ah   Ax=8F7Ah CF=0, SF=1
ADD Ax, 7AF8h
Mov Bx, 0FA77h
INC Bx

```
 1 1 1
 8F7A
 7AF8
-----
10A72
```

Ax=10A72h, CF=1, SF=0
Bx= FA77h CF=0, SF=1
Bx = FA78h, CF=0, SF=1.

Q#5.

(i) The value is assigned to these symbolic Constants b/c EQU directive is used.

(ii) MOV EAx, second
IMUL EAx, minutes
IMUL EAx, Hours
IMUL EAx Days

(iii) mov [secondsInWeek], EAx
call WriteInt

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Q#6:- Include Irvine 32.inc

```
.data
array word 10,20,30,40,50,60,70,80,90,100.

.code
main PROC
mov ESI, offset array
mov ax, WORD PTR[ESI]
 lea ebx, [ESI+2]
 mov ecx, 9
 loop:
   mov dx, WORD PTR[ebx]
   mov WORD, PTR[ESI],dx
    add ESI,2
   add ebx,2
   dec ecx
   jnz loop.
  mov WORD PTR[ESI],ax
   mov ecx,10
   mov esi, offset array
print loop:
     mov ax,[ESI]
   movzx eax,ax
    call writeDec
    call Crlf
    add esi,2
    loop printloop.
exit
main ENDP
 END main.
```

Output:
20
30
40
50
60
70
80
90
100
10

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Q#7.

0xA1B2C3D4 32 bit
memory address

(a)	Little Endian	(b)	Big-Endian
3000h	D4	3000h	A1
3001h	C3	3001h	B2
3002h	B2	3002h	C3
3003h	A1	3003h	D4

(c) Mov, EAX, [3000h]

or

mov EAx, 0A1B2C3D4.

Q#8.

Registers (EAx, EBx, ECx, EDx) can be split into small registers (Ax, AH, AL)

The values taking 8 bits is more efficient & convinient to be stored in AH or AL instead of Ax or EAx.

Registers are splited for flexibility & compatibility.

Ex:- mov AL, 'A' is more efficient than mov EAx, 'A' wasting extra 24 bits

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Q#3,

· A: (A+B) - (C+D)
using EAx, EBx, ECx, EDx & show the steps.

```
.data
A DWORD ?
B DWORD ?
C DWORD ?
D DWORD ?

.code
mov EAx, A
mov EBx, B
mov ECx, C
mov EDx, D

add, eax, ebx      A+B
add ecx, edx       C+D
sub eax, ecx       (A+B)-(C+D)
mov A, eax
call WriteInt
exit
```

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Q#10:

(i) One to One relationship:-
A record in one entity is associated with exactly one record in another entity.

→ One to many relationship:-
A record in one entity is associated with many records in another entity.

(ii) Memory access takes more time as compare to the registers because registers are present in CPU where memory storage unit is external & to access that buses involvement is necessary which increase the overall time required to process.

(iii) Machine dependent languages:-
Those languages which depends on the architecture & works on specific architecture
Ex:- Assembly language for intel x86 can't run on ARM processors.

Machine Independent languages:- Can run on multiple platforms without rewriting.

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(iv) Computer Organization: How hardware is built comes under Computer Organization means how hardware is connected & Operated.

Whereas

Computer Architecture is associated with what it can do (the set of instruction, addressing modes & design concepts).

(v) Carry flag (CF) set when an unsigned arithmetic operation produces a carry / borrow.

(if signed then Overflow flag is set (used))

(vi) Source file contains the source code (.asm file) where as listing file contains the metadata, machine code & errors of the source file when passed through the Assembler.

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(vii) Memory reading take longer than reading from register because it involves the buses which usually increase time because it perform transfering of data, fetching of data, approval of data etc.

(viii) Real mode:
20 bit address
1MB memory usage
→ No protection for illegal memory access

Protected Mode:
32/64 bit
4GB memory
Multitasking.

→ used in MS DOS system.

→ Real mode is not for modern OS because it is too limited, 1MB max memory, no process isolation.

(ix) Essential Components are buses which includes:
(i) Address Bus: carries memory address
(ii) Control Bus: signals whether it's a read/write
(iii) Data Bus: transfers actual data between CPU & memory